Works cited:
https://www.hopkinsmedicine.org/health/wellness-and-prevention/the-science-of-sleep-understanding-what-happens-when-you-sleep
https://www.ninds.nih.gov/health-information/public-education/brain-basics/brain-basics-understanding-sleep
https://www.sleepyhead.co.nz/sleep-guide/science-behind-sleep-cycles/
https://picower.mit.edu/news/sleeping-brain
https://sleep.hms.harvard.edu/education-training/public-education/sleep-and-health-education-program/sleep-health-education-47
https://magazine.ucsf.edu/why-cant-i-sleep

During sleep: will cycle repeatedly through 2 different types of sleep

- REM (rapid eye movement) sleep & non REM

Cycle:



SLEEP CHART STAGE

	Stage 1 - falling asleep	Stage 2 - light sleep	**Stage 3 -** deep sleep	**REM SLEEP**
Duration	several minutes, 5%	Spend more of your repeated sleep cycles in stage 2 than in other stages, 45%	longer periods during the first half of the night, 25%	Gets longer after each cycle, 25%
Body movements	Heartbeat, breathing, & eye movements slow, & muscles relax with	Heartbeat & breathing slow, & muscles relax even further, body temperature 🔽 & eye	Heartbeat & breathing slow to their 🔽 levels during sleep, muscles are relaxed and it may	Eyes move rapidly behind closed lids, breath rate increases + body becomes

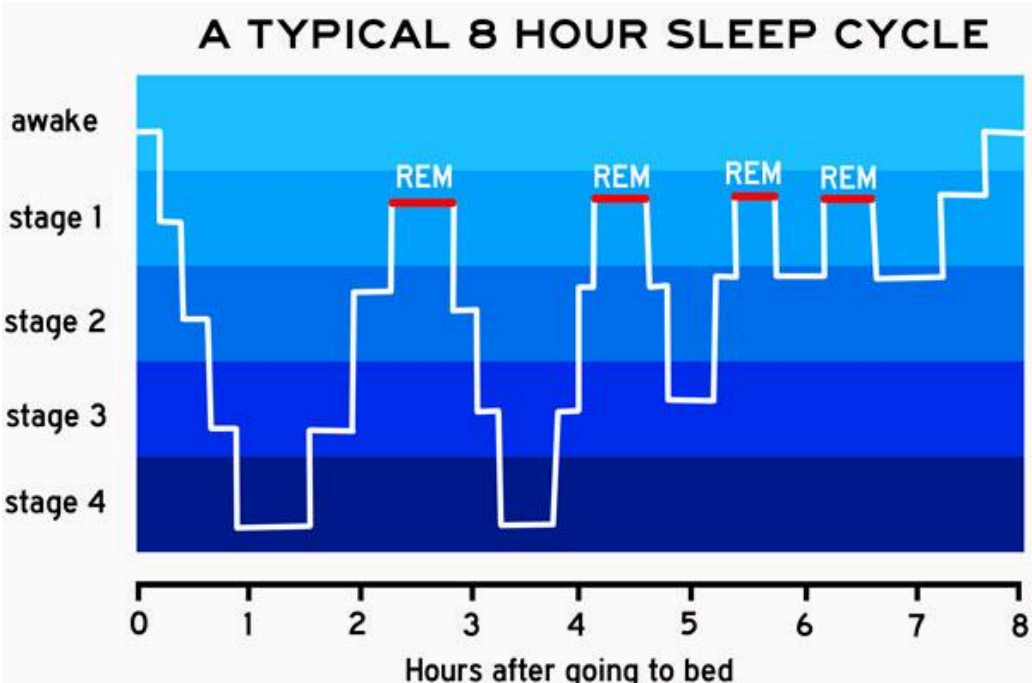
	occasional twitches	movements stop	be difficult to awaken you	temporarily paralyzed during dream (stop from hurting ourselves)
Brain waves (produced by large networks of neurons firing in synchronicity)	Brain waves begin to slow	Brain wave activity slows but is marked by brief bursts of electrical activity	Brain waves become even slower, strong	Brain waves are similar to those during wakefulness
Importance	Transitory stage from awake —> asleep	Brain organizing memories and information from the time you spent awake, filter out sensory information	Learning and memory, most restful and restorative, repair injuries and reinforce your immune system, metabolic regulation, hormone release	Memory consolidation, learning, and creativity

REM SLEEP (more info):

- **Hippocampus:** strengthens memories by replaying the day's experiences, processing, categorising them before moving them into long-term memory
- **Amygdala:** processes emotional memories, tagging feelings, prioritising them for storage
- Emotions are more likely to be processed apart from factual content, and so less likely to be filtered
    - By replaying emotional moments while dreaming, we're able to experience them with less intensity, so we can deal with them in a more balanced way.

Cycle 🔁 ~4-5 times, 90 to 120 minutes each:

- stage 2 NREM sleep: each one is longer than the last
- Each time more REM, less deep sleep



Awake



REM



Non-REM stage 1



Non-REM stage 2



Non-REM stage 3, “slow wave”



WAVES:

Non-REM sleep:

- Amplitude: ⬆️
- Frequency: Slow
- Small chunks of experience are replayed at high speed

REM sleep (brought by 📈 of brain chemical acetylcholine in brain’s outer surface)

- Amplitude: ⬇️ (higher than wakefulness)
- Frequency: Fast (slower than wakefulness)
- Alpha rhythm
- Memories are replayed in real-time and sensations like motion are rekindled

NEURONS:

- 📉 in the activation or "firing" rate of most neurons throughout the brain (wakefulness —> non-REM sleep)
- Wakefulness: random & variable activity pattern
- Non-REM sleep: coordinated & synchronous pattern
- REM sleep: random and variable (similar to wakefulness)
    - 📈 in the firing rate of most neurons throughout the brain, as compared to non-REM sleep

TEMPERATURE:

- Before we fall asleep: Lose some heat to the environment
    - Help induce sleep

- During sleep: Central set temperature is reduced by 1-2°F
    - Use less energy maintaining our body temperature
    - Conserve energy
- REM sleep: body temperature falls to lowest point

RESPIRATORY CHANGES:

- Awake: breathing is irregular
    - affected by speech, emotions, exercise, posture, etc.
- Wakefulness-stages of non-REM sleep: breathing rate slightly 📉 and becomes very regular
- REM sleep: breathing rate 📈 overall
    - Pattern becomes much more variable again

CARDIOVASCULAR ACTIVITY (Heart):

- Function of sleep: Give the heart a chance to rest from the constant demands of waking life
- Non-REM sleep: overall 📉 in heart rate & blood pressure
- REM sleep: overall 📈 in blood pressure & heart rate, variation in cardiovascular activity
    - Changes in blood flow:
        - Cause erections to occur in males/swelling of the clitoris in females

INCREASED PHYSIOLOGICAL ACTIVITY DURING SLEEP:.

- Overall, many physiological activities are reduced during sleep
    - Ex. Kidney function slows & production of urine 📉
- Some physiological processes may be maintained/even increased during sleep
    - Ex. 📈 in the release of growth hormone & physiological activities associated with digestion, cell repair, and growth are often greatest during sleep, suggesting that cell repair and growth may be an important function of sleep

Causes for sleep:

Sleep pressure:

- Inevitable accumulation of sleepiness each day
    - Caused by a gradual buildup of the molecule adenosine in 🧠
    - Adenosine is a byproduct of expending energy; it's freed up from the energy-carrying molecule ATP, which fuels all living cells

Circadian rhythms:

- 24-hour biological 🕐 keeps time for your every cellular function, allowing your body to reliably anticipate and respond to its physiological needs (ex. sleep)
    - Proteins build up each night and degrade each day, and the whole system is reliant on environmental cues, most notably natural light

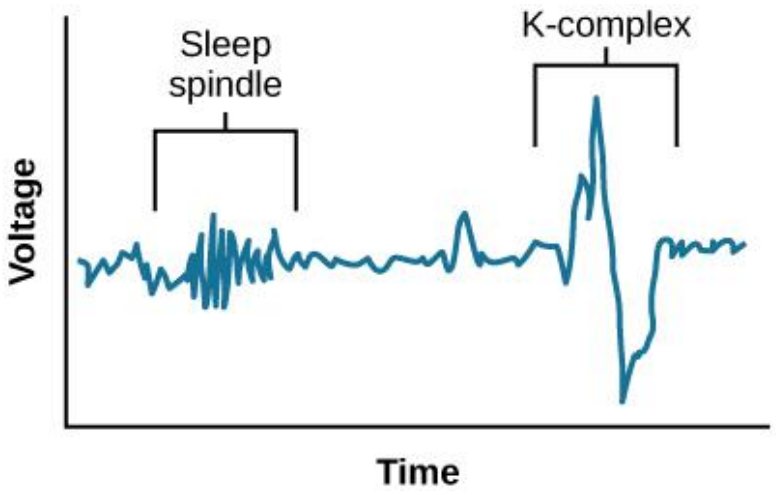
STAGE 2 NREM SLEEP:

Sleep spindle:

- brief, rhythmic bursts of brainwave activity (12-14 Hz) that appear as "spindle" shapes on an EEG during Stage 2 NREM sleep
- crucial for processing + consolidating memories, learning new skills, maintaining sleep stability
- plays a role in emotional regulation to recovery from stress

K-complexes:

- single, high-amplitude waves
- Can be triggered by external or internal stimuli & may help protect sleep and contribute to memory consolidation



THALAMUS WTFFFFFFFF

- network of cell groups activate the thalamus & cerebral cortex
- key switch in the hypothalamus shuts off this arousal system during sleep
    - other hypothalamic neurons stabilize the switch (absence —> inappropriate switching of behavioural states —–> narcolepsy, insomnia, etc.)
- acts as a gatekeeper for sleep, blocking sensory information during NREM sleep while relaying it during REM sleep for dreams
- Crucial for regulating sleep cycles by controlling thalamo-cortical rhythms (ex. delta waves, sleep spindles)

Works cited:
https://my.clevelandclinic.org/health/body/12148-sleep-basics
https://my.clevelandclinic.org/health/body/12148-sleep-basics
https://medlineplus.gov/sleepdisorders.html
https://www.ohsu.edu/brain-institute/understanding-sleep-disorders
https://sleep.hms.harvard.edu/education-training/public-education/sleep-and-health-education-program/sleep-health-education-20

Common disorders:

- Central disorders of hypersomnolence
    - ex. narcolepsy
        - Severe daytime sleepiness, dreamlike hallucinations while falling asleep or waking up, temporary muscle weakness
- Circadian rhythm sleep-wake disorders (ex. jet lag or shift work sleep disorder)
    - problems with the sleep-wake cycle (make you unable to sleep and wake at the right times)
- Insomnia
- Parasomnias
    - acting in unusual ways while falling asleep, sleeping, or waking from sleep, such as walking, talking, or eating
    - sleepwalking: This occurs when the parts of your brain that control walking and physical activities stay active while you sleep
        - sleep terrors (aka night terrors) - form of sleepwalking may cause people to scream, break into a sweat or get out of bed abruptly
- Sleep-disordered breathing
    - ex. sleep apnea (a breathing disorder in which you stop breathing for 10 seconds or more)
    - Occurs when the soft tissues in your throat block your upper airway, cutting off your oxygen supply for a few seconds
- Sleep-related movement disorders
    - ex. restless leg syndrome (a tingling or prickly sensation in your legs, along with a powerful urge to move them)

Stages 1 & 2 - NREM sleep

- Sleep starts or twitches (sleep myoclonus)
- Exploding head syndrome
- Teeth grinding (bruxism)
    - Common among heavy drinkers, smokers, people who are under stress
- Periodic limb movement disorder (PLMD)

Stage 3 - NREM sleep

- Night terrors (also known as sleep terrors)
- Sleepwalking (somnambulism)
- Confusional arousals (partial waking with sleep inertia)
- Periodic limb movement disorder (PLMD)

- Sleep-related eating disorder

REM sleep

- Nightmares and nightmare disorder
- REM sleep behavior disorder
- Hypnagogic and hypnopompic hallucination

Insomnia: disruptions in how you feel/function because you aren't sleeping well/enough

Duration:

- Acute (short term)
- Chronic (long term)

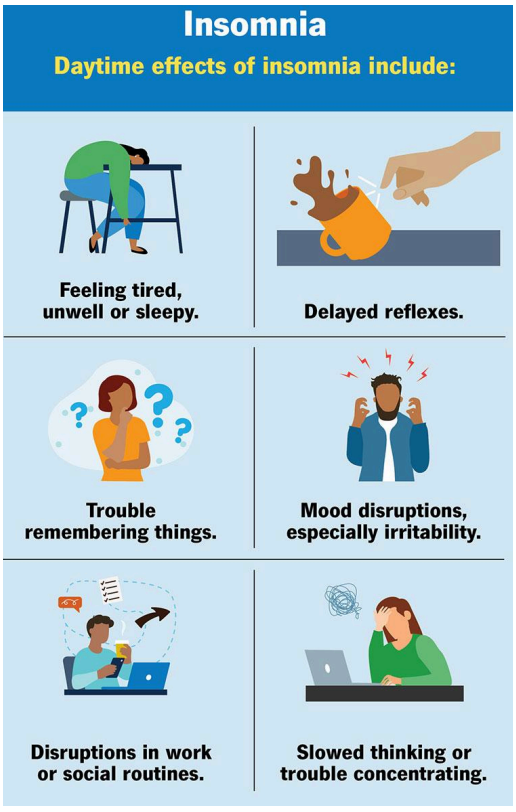
Cause:

- Primary - happens on its own
- Secondary - symptom of another condition or circumstance
- Hyperarousal
    - People are physiologically & mentally activated
- increased heart rates, elevations in the stress hormone (cortisol), greater amounts of fast brain activity around sleep onset

Types:

- Initial (sleep onset) insomnia: Trouble falling asleep
- Middle (maintenance) insomnia: Wake up in the middle of the night but you fall back asleep **(most common)**
- Late (early waking) insomnia: Wake up too early in the morning and don't fall back asleep

Symptoms:



- Long periods of not breathing, which may indicate apnea
- Excessive daytime sleepiness
- Deep snoring, noise or restlessness while sleeping
- Irregular breathing or increased movement during sleep
- Difficulty falling asleep
- Headaches upon waking
- Consistent nightmares
- Physical reactions to dreams

Causes:

- Family history (genetics)
    - Sleep traits and conditions run in 👪
- **Brain activity differences**
    - May have more active 🧠s or brain chemistry differences that affect their ability to sleep
- Medical conditions
    - Physical health
        - ex. temporary illnesses like minor infections/injuries, or chronic conditions like acid reflux or Parkinson's disease
        - conditions that affect your circadian rhythm, your body's natural sleep/wake clock
    - Mental health
        - ex. anxiety/depression
- Life circumstances
    - Stressful/difficult life circumstances may not necessarily cause insomnia, but can contribute to it
- Life changes
    - Brief or temporary changes
        - ex. jet lag, sleeping in an unfamiliar place or adjusting to a new work schedule (especially shift work), moving to a new home
- Your habits and routine
    - Sleep habits (also known as sleep hygiene)
        - ex. Whether or not you take naps, when you go to sleep, if and when you consume caffeine

Works cited:
https://www.mayoclinic.org/healthy-lifestyle/adult-health/in-depth/sleep/art-20048379
https://www.nhs.uk/every-mind-matters/mental-wellbeing-tips/how-to-fall-asleep-faster-and-sleep-better/
https://www.health.harvard.edu/newsletter_article/8-secrets-to-a-good-nights-sleep
https://www.healthline.com/nutrition/17-tips-to-sleep-better
https://my.clevelandclinic.org/health/treatments/15308-sleeping-pills
https://www.medparkhospital.com/en-US/lifestyles/sleeping-pills
https://www.hopkinsmedicine.org/health/wellness-and-prevention/sleep-aids

Sleeping Pills:
- Can cause drowsiness or silence the area of your 🧠 that keeps you alert

Over the counter sleep aids:
- Most OTC sleeping pills contain antihistamines
    - Antihistamines work by blocking histamines (substances in your body that cause alertness)
    - Antihistamines –> diphenhydramine + doxylamine (sedating antihistamines)

Natural sleep aids (supplements):
- Melatonin - hormone that promotes sleep (can influence when you fall asleep/wake up)
    - Pineal gland secretes melatonin —-> sleep & responsible for the sleep cycle
    - Secreted at night/low-light conditions to aid body's relaxation & help people sleep
    - Comes in short-acting & long-acting forms
- Valerian - natural calming effect on your nervous system & helps to promote better sleep

Prescription sleep aids:
- Benzodiazepines - stimulates a chemical in your 🧠 called GABA
    - Leads to sedation, 💪 relaxation, reduced anxiety, improved learning & memory
    - 📈levels of brain GABA neurotransmitters
- Z-drugs work by slowing the activity in your brain (fast-acting)
        - Induce sleepiness, calm anxiety, & promote sound sleep without leaving you feeling groggy/lightheaded in the morning

Tips:

1. Stick to a sleep schedule = good sleep hygiene
    - Set aside no more than 8 hours for sleep (recommended sleep for a healthy adult is at least 7 hours)
    - Go to bed and get up at the same time every day
        - Helps reinforce your body's sleep-wake cycle
    - ⭐If not asleep in ~20 minutes of going to bed, leave your bedroom and do something relaxing (ex. Read or listen to soothing music) Go back to bed when you're tired. Repeat as needed, but continue to maintain your sleep schedule and wake-up time

2. Pay attention to what you eat and drink
    - Don't go to bed hungry/stuffed

        - Avoid heavy meals within a couple of hours of bedtime (can cause discomfort)
    - Avoid nicotine, caffeine, and alcohol (stimulates that can make us more alert)

3. Create a restful environment
    - Keep your room cool (~65 F), dark, and quiet
        - Avoid exposure to light + light-emitting screens before bedtime
        - Ex. room-darkening shades, earplugs, a fan, ambient noises, good bedding
    - Do calming activities before bedtime (ex. bath, relaxation techniques)
    - 📈bright light exposure; 📉blue light exposure

4. Limit daytime naps
    - Limit naps to no more than 1 hour and avoid napping late in the day
    - Exception for people with night shifts
    - Try to reserve bed to only sleep (stimulus for sleeping, not wakefulness)

5. Include physical activity in your daily routine
    - Regular physical activity
    - Avoid being active too close to bedtime - can increase alertness & levels of hormones (ex. epinephrine and adrenaline)

6. Manage worries
    - Try to resolve your worries/concerns before bedtime
        - Solution: Jot down what's on your mind and then set it aside for tomorrow.
        - Stress management: get organized, set priorities,delegate tasks, meditate
    - Know when to contact your health care provider

7. Take Melatonin supplement
    - Melatonin: key sleep hormone that tells 🧠 when it’s time to relax and head to bed

THE BIG DOGS 🐺

- Somnee
- Muse
    - myndlift
- Mendi

https://neuromodec.org/what-is-transcranial-alternating-current-stimulation-tacs/
https://pmc.ncbi.nlm.nih.gov/articles/PMC7867505/
https://www.sciencedirect.com/science/article/pii/S1364661322002984
https://www.peakbraininstitute.com/

Mapping
Create a working model of your brainprint using EEG+ and proprietary AI/ML that decode your brain's unique, sleep-promoting neuronal patterns

EEG (electroencephalogram AKA electric brain graph)

- painless brain test that records your brain's electrical activity
- activity can be amplified, analyzed, and visualized
- EEG signal is sensitive to external noises & requires delicate handling.

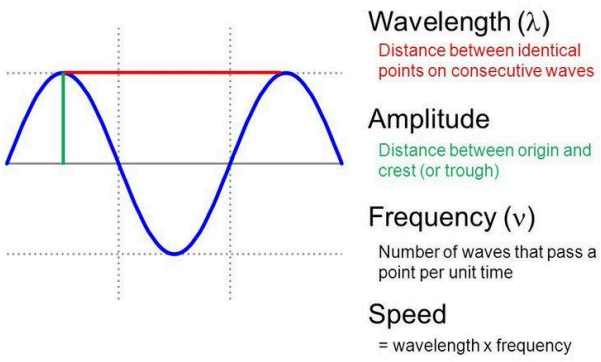
- brainwaves convey information in rate of repetition
    - Oscillations (measured on scalp), aka frequencies, are measured as Hz (hertz)

Slow activity (lower frequency and high amplitude)

- Think: low, the deep beat of a drum

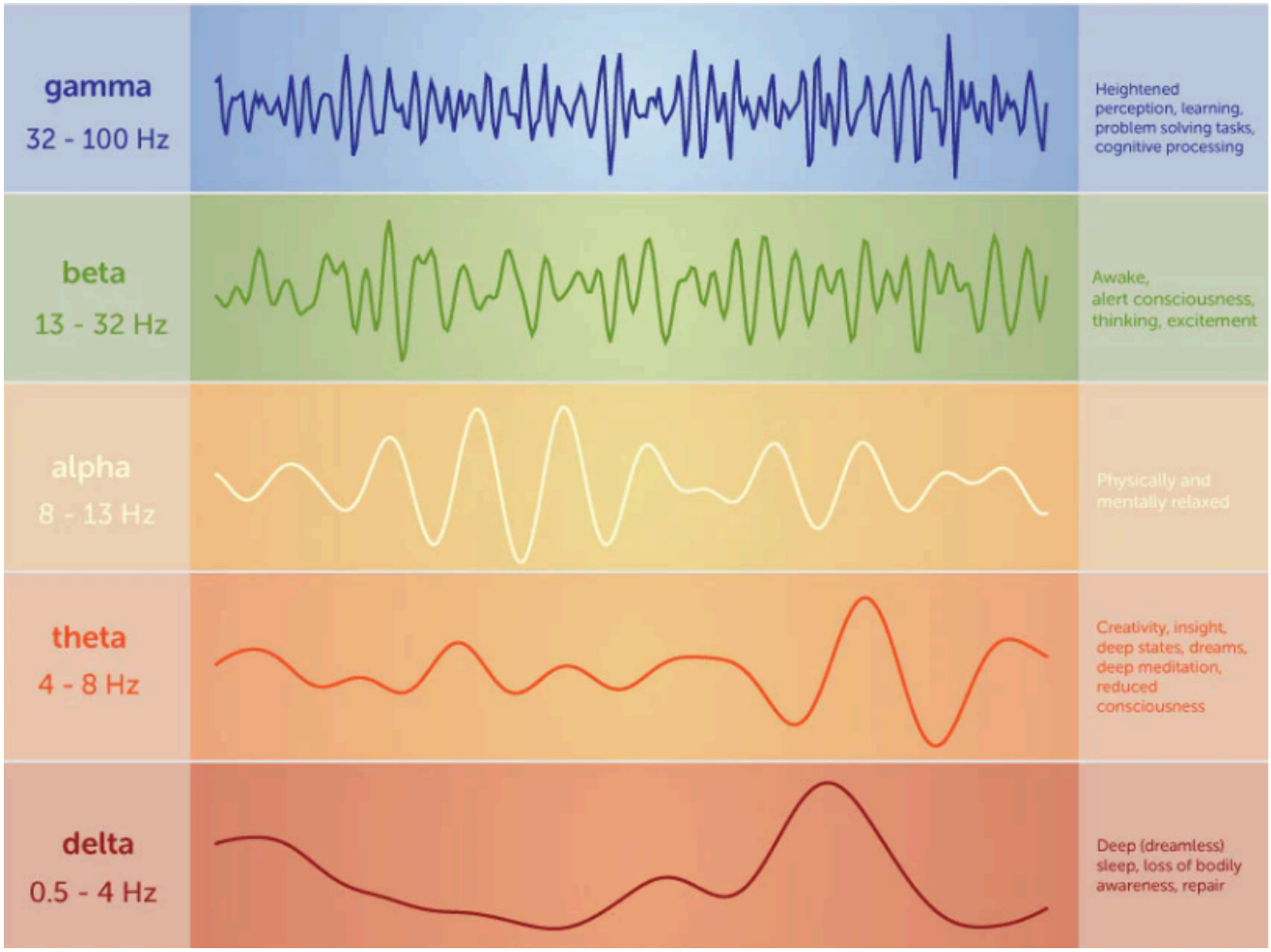
Fast activity (higher frequency & often smaller amplitude)

- Think: high pitched flute
- High frequency = greater alertness



Neurofeedback with MUSE + Myndlift:

- During neurofeedback training - user is rewarded with positive visual and/or auditory feedback when specific brainwaves are in the target range.
    - ex. relaxation training: when brainwaves reflect a relaxed state, a rewarding sound is presented
    - Conditioning reinforces the desired state over time



Gamma:

- peak mental state - simultaneous processing of information from different parts of the 🧠

Beta:

- ex. active conversation, making decisions, solving a problem, focusing on a task, learning a new concept
- ***easiest to detect when we're busy thinking actively

Alpha:

- ***some of the most easily observed & were the first to be discovered
- become detectable when 👀 are closed & the mind is relaxed
- ex. Yoga, just before falling asleep, being creative and artistic

Theta:

- indicate deep relaxation & occur more frequently in highly experienced meditation practitioners

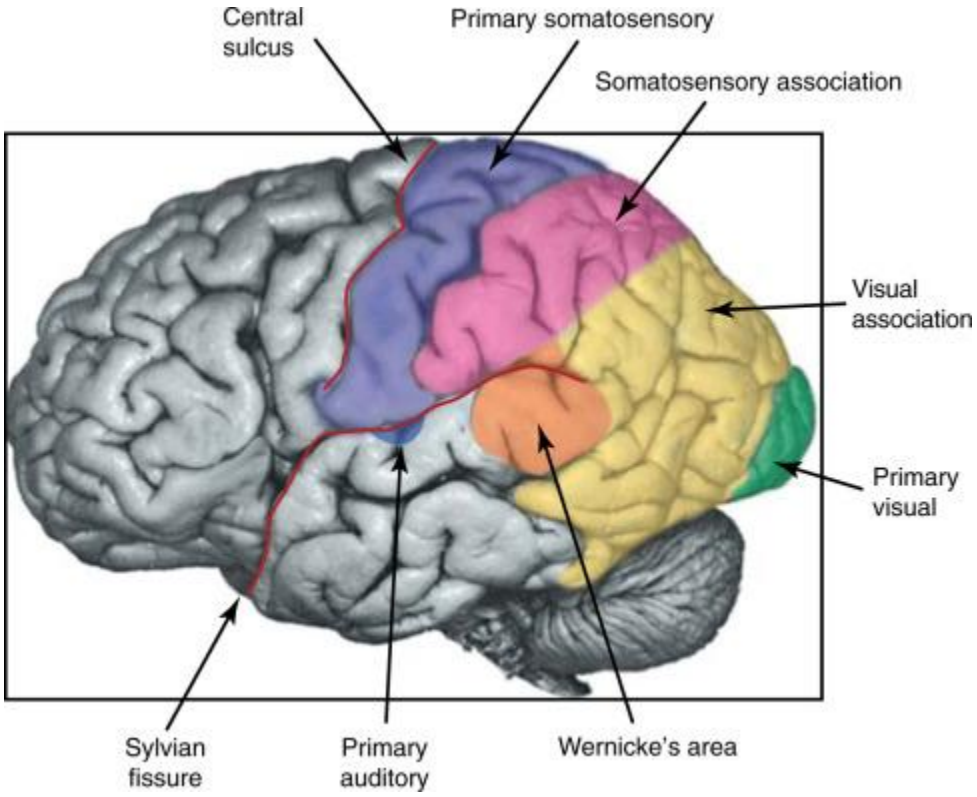
    - source: frontal parts of the brain (associated with monitoring of other mental processes)
- positive association of theta waves with memory, creativity, & psychological well-being
- strongly detectable when we're dreaming in our sleep
- ex. Deep meditation, daydreaming, doing a task that is so automatic that the mind can disengage from it (brushing teeth, showering)

Delta:

- ***slowest of all brainwaves
- strongest when we are enjoying restorative sleep in a dreamless state
- state where healing & rejuvenation are stimulated

WHERE:

- Sleep regulation - on sensorimotor strip (aka C-strip)



- C-strip generates:
    - Sleep spindle production
    - SMR (sensorimotor rhythm) regulation
        - *def.* specific brainwave frequency (around 12-15 Hz) generated in the sensorimotor cortex (brain area for movement), representing a state of relaxed alertness, stillness, and focused attention
    - Sharp wave complexes - SMR spindle after a k-complex, to keep us asleep at
    - Beta rhythm management - to manage vigilance + keep us asleep
    - Sleep onset control - body based low Beta and SMR (12-15hz range) at the vertex

fNIRS neurofeedback (MENDI STRAT) - functional near-infrared spectroscopy

- detects changes in the 🧠 local blood flow & oxygenation in response to neural activation
- fNIRS sensor is placed over the forehead, beneath it is the prefrontal cortex (PFC)
    - PFC plays a key role in the mental processes of the Executive Function (e.g. focus and working memory)
- fNIRS is similar to 🏋️ training for the PFC - with repeated training lasting structural and functional neuroplasticity (*def. ability of 🧠 to change its structure & function throughout life in response to experiences, learning, or injury)* will occur
- focus: 🧠 neuroplasticity (ability to reorganize itself by forming new neural connections throughout life - strengthen beneficial patterns and weaken those that contribute to dysfunction)
    - Neurofeedback tells the 🧠 what neural pathways are positive

URGOnight strat

- focuses on sensorimotor-rhythm (SMR) waves (low-beta waves)
    - 12-15 hertz range, calm, focused state, like "practice energy" for learning
- reward: little pings/animations when they produce waves in that range
- cues help you understand how to produce more SMR waves when awake —-> greater production of *sleep spindles* while asleep
    - *def.* (brief squiggles in the 10-15 hertz range that are thought to be associated with memory processing during sleep)
- Tips on how to produce more SMR brainwaves: focusing on breathing, clearing the mind, thinking positive thoughts, etc.

Elemind strat:

- 1 Hz dominate brain activity in deepest stages of sleep
    - Important for physical recovery, clearance of waste products in brain, rejuvenation of tissues

Brain peak Institute Strat:

- Approach:
    - C-strip regulation for sleep control
    - Heart rate variability training before neurofeedback sessions
    - 🧠 wave optimization: addressing amplitude & coherence features that are keeping the C + F areas stuck
    - Sleep pattern regulation - repeating SMR training & Beta training, daily sleep reports to enable adjusting
    - Training for sleep maintenance (C3 Beta) & sleep onset (Cz SMR) resources

tACS (Transcranial Alternating Current Stimulation)

⚡

Other implementations:

Heart rate variability (HRV) biofeedback: trains individuals to regulate their heart rate

- promotes a state of calmness & can be useful before bed as it prepares the body for rest
- Stress & anxiety —-> insomnia (= 📈 heartbeat & restlessness)
    - HRV biofeedback can help balance the autonomic nervous system

Electromyography (EMG) biofeedback: can help reduce muscle tension

- Muscle tensions —> restlessness & difficulty falling asleep

Galvanic skin response (GSR) biofeedback: measures changes in sweat gland activity

- sweat gland activity = stress levels

Breathing biofeedback: helps regulate the nervous system through guided slow breathing exercises

Issues:

- Neurofeedback sleep bc outside noises (like blinking), scalp, etc. can effect measurement of brainwaves from the “band”
- Lack of research/correlation between neurofeedback and betterment of sleep

Best method: Cognitive behavioral therapy for insomnia (CBT-I)

- 🧠 essentially seesaws between its sleep and wake system
- wake side of the seesaw in insomniacs is heavily weighed down, CBT-I rebalances seesaw
- Uses: deliberate sleep restriction to weigh down the sleep side, teaches patients how to build constructive thoughts around sleep, uses calming exercises to lighten the stress side

Peak Brain Institute

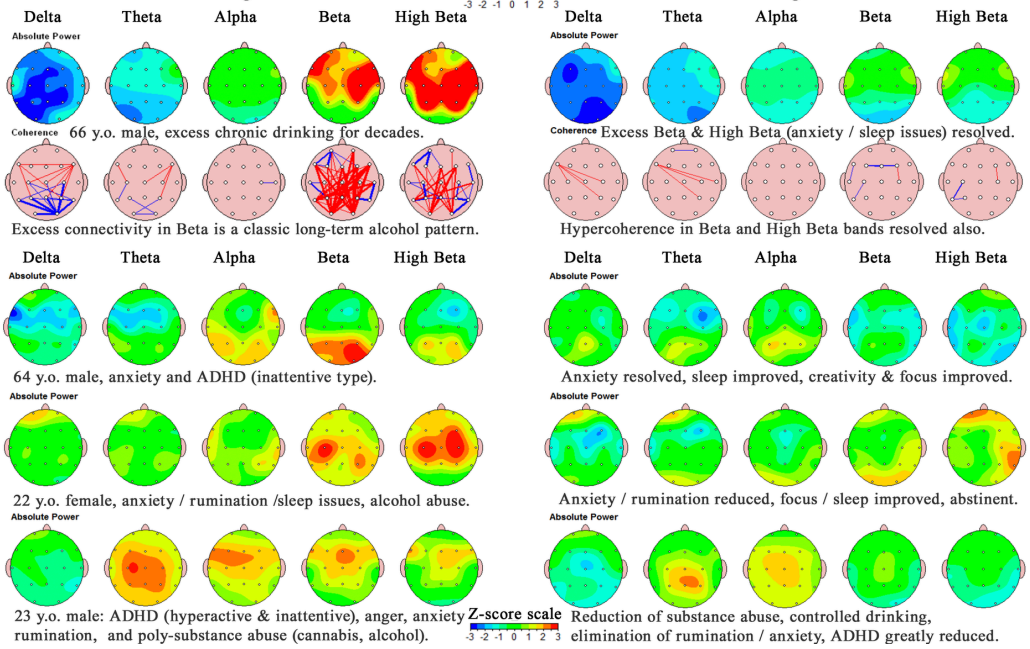
QEEG Z-scored "Brain Maps" Pre- and Post-Neurofeedback (avg 30 sessions).

Pre-training Baseline

Z-score scale

-3 -2 -1 0 1 2 3

Post-training EEG Results



66 y.o. male, excess chronic drinking for decades.

Excess Beta & High Beta (anxiety / sleep issues) resolved.

Excess connectivity in Beta is a classic long-term alcohol pattern.

Hypercoherence in Beta and High Beta bands resolved also.

64 y.o. male, anxiety and ADHD (inattentive type).

Anxiety resolved, sleep improved, creativity & focus improved.

22 y.o. female, anxiety / rumination /sleep issues, alcohol abuse.

Anxiety / rumination reduced, focus / sleep improved, abstinent.

23 y.o. male: ADHD (hyperactive & inattentive), anger, anxiety rumination, and poly-substance abuse (cannabis, alcohol).

Z-score scale

-3 -2 -1 0 1 2 3

Reduction of substance abuse, controlled drinking, elimination of rumination / anxiety, ADHD greatly reduced.

-

Basics / What tACS Is

- tACS = transcranial alternating current stimulation, non-invasive (nothing goes inside the 🧠)
- Arrangement of electrodes: one electrode on target brain area + reference somewhere neutral
- uses low-intensity sinusoidal AC through electrodes on scalp
    - current passes through skull 💀 → affects surface cortical neurons
    - AC = current flips positive → negative every half cycle
    - current flows anode→cathode, then flips direction next half
- goal = sync with brain's natural electrical rhythms ("oscillations")
- usually 1 frequency <100 Hz

🧠 Concept / Why tACS Works

- brain produces rhythmic electrical activity = brain oscillations
- measured by:
    - EEG = electroencephalography → picks up electrical signals at scalp
    - MEG = magnetoencephalography → picks up magnetic fields from brain activity
- different frequencies = different brain functions (ex. alpha = relaxation, theta = memory)
- brain regions communicate by syncing their rhythms
- tACS tries to sync (couple) or unsync (decouple) those rhythms
- tACS doesn't force neurons to fire → instead changes the timing of when they're likely to fire (subthreshold)

🧩 Biophysics (Simplified)

- typical intensity: 1–4 mA → creates ~0.1–1 mV/mm electric fields in brain
- these fields are small but similar in size to normal brain activity
- effects depend on:
    - where electrodes are placed
    - how conductive brain tissues are
    - the shape of your head
    - orientation of neurons
- pyramidal neurons (layer V) = most affected
    - long shape → extra sensitive to weak fields
- entrainment = brain rhythm "locks onto" the external tACS rhythm
- Arnold tongue = stronger stimulation → can entrain a wider range of frequencies
- can entrain harmonics = multiples (2×) or fractions (½×) of brain's natural frequency
- threshold for entrainment ≈ 0.3–0.4 mV/mm

🔄 Mechanisms (Easy Explanation)

1. Stochastic resonance

    a. neurons respond randomly but get nudged by tACS → no strict “minimum power needed”
2. Rhythm resonance
    a. best effects happen when stim frequency = brain’s own frequency.
3. Temporal spike biasing
    a. tACS changes when neurons fire, not how many fire
4. Network entrainment
    a. whole circuits start syncing with the stimulation.
5. Imposed pattern
    a. if stim is strong enough, it can override normal rhythms and create a new one

⏱️ Online vs Offline Effects

➡️ online = effects DURING stimulation

- brain responds instantly while current is on
- Includes:
    - entrainment (syncing to tACS freq)
    - spike timing changes
    - increased or decreased oscillation power
    - short-term connectivity changes

➡️ offline = effects AFTER stimulation ends

- these come from neuroplasticity (brain physically changing connections)
- based on STDP = spike-timing dependent plasticity
    - LTP = stronger connection (when neuron A fires → then B fires)
    - LTD = weaker connection (B fires before A)
- offline changes can last ~70 minutes after a 20-minute session

📡 Neuronal-Level Effects

- current makes neuron membranes slightly oscillate → small depolarizations/hyperpolarizations
- affects probability of firing, not firing rate
- neurons fire more at preferred phase of the sine wave (phase-locking)
- tACS can affect:
    - Bursting, timing patterns
    - rhythm stability, local + long-range synchronization

🕸️ Network-Level Effects

- tACS can change:
    - local oscillation power (how strong a rhythm is)
    - phase connectivity (how rhythms line up between regions)
    - coherence (how well two regions talk to each other)
- can increase or decrease communication between brain areas
- effects depend heavily on brain state (awake vs tired, task vs resting)

🧪 Comparison to Other Brain Stimulation

🔋 tDCS vs tACS (both are subthreshold)

- tDCS = constant current (just shifts excitability up/down)
- tACS = rhythmic current (targets timing & oscillations)

tACS = better for modulating brain rhythms specifically

🧲 TMS vs tACS

- TMS uses a magnetic pulse → induces electric current → fires neurons directly
- tACS gently nudges timing, doesn't cause firing
- TMS = strong, used clinically
- tACS = cheap, portable, safe, mild

tACS good for reshaping rhythms without heavy stimulation load

🧠💡 Cognitive Effects & Applications

- can modulate:
    - working memory
    - long-term memory, attention
    - Perception, learning & decision-making
- Strategies:
    - boost oscillation power in target areas (ex: alpha for relaxation/sleep)
    - modulate theta–gamma coupling (important for memory)
    - synchronize or desync distant brain regions
    - adjust trade-offs: fast vs accurate, stable vs flexible thinking
    - frequency choice matters A LOT
    - individualized (using your own peak frequency) often works better
    - multi-electrode setups allow phase-control between brain regions

[Beacon Biosignals]

Example design - beacon biosignals

Battery

- Maximum charging time: 3 hours and 30 minutes
- Average charging time: 1 hour of charging for up to 8 hours of recording

Storage

- Maximum number of recordings on memory: 24 hours
- Maximum recording duration: 24 hours

Output

- EEG Channels: 5
- Sleep endpoints: TST, WASO, N1, N2, N3, REM sleep, SOL (Space-Occupying Lesion - referring to any abnormal mass or collection, ex. tumor that takes up space within the skull or spinal canal), number and duration of awakenings
- Raw data: EDF



EEG sensors:

A. 2 frontal sensors in F7 and F8 locations to measure frontal brain activity
B. 1 ground electrode on the frontal band (Fp1 Location)
C. 2 occipital sensors on O1 and O2 locations to monitor occipital brain activity

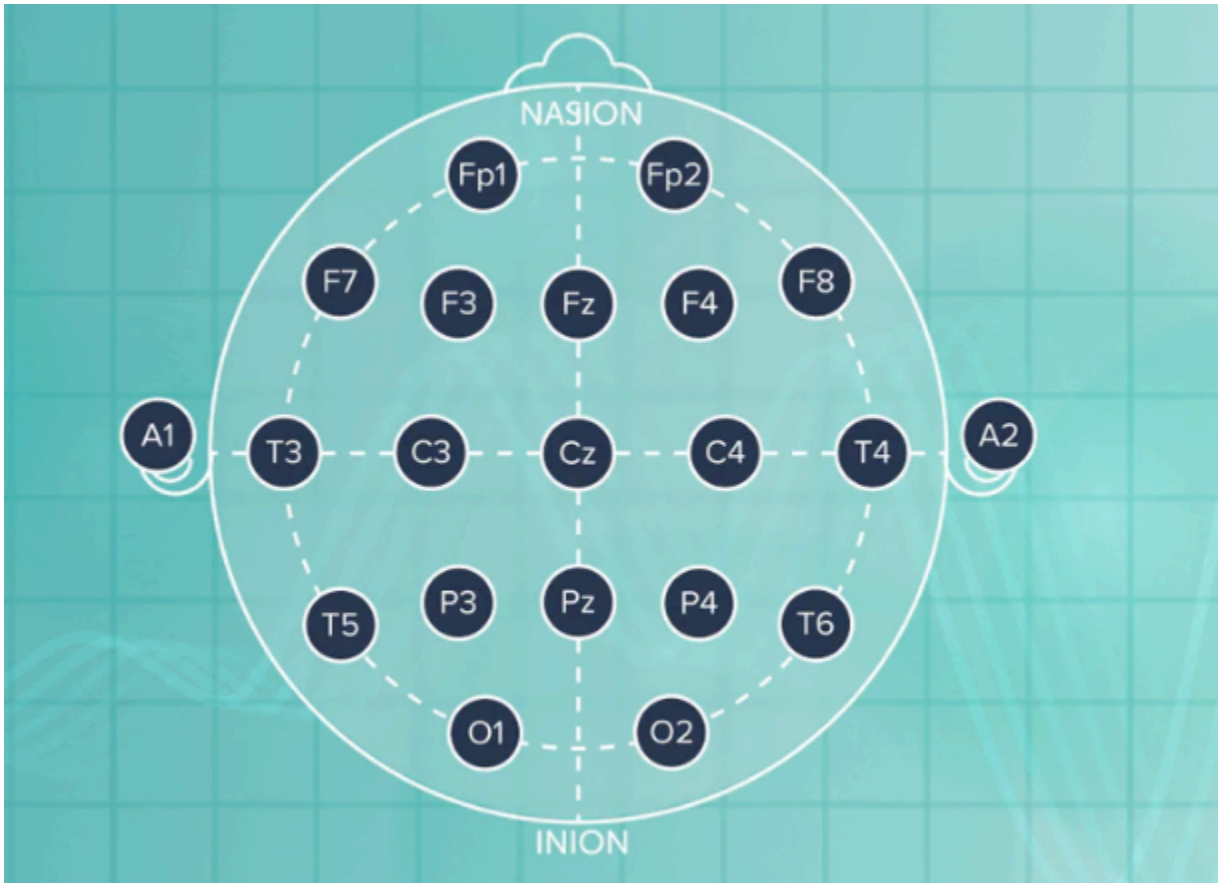
Accelerometer:

D. 3D accelerometer to measure movements, head position, and respiratory rate/trace during sleep

Power

E. Power button
F. Magnetic port for charging

Muse



EEG Sensors:

- 2 on the forehead (FP1, FP2)
- 2 behind the ears (T9, T10)
- C strip coordinates
- reference sensor (FPz) - detect and measure the activity of your brain
    - PPG sensor measures your heart rate, while an accelerometer and gyroscope measure your breath rhythm and body movement